



Birla Vishvakarma Mahavidyalaya Engineering College

(An Autonomous Institution)

Managed by Charutar Vidya Mandal

IT Department



Mid-1 Presentation

Mini Project (3IT31)

Academic Year 2025-26

Face Recognition Based Attendance System

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Batch:

D10

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The Vulnerability of Traditional Attendance

The Problem

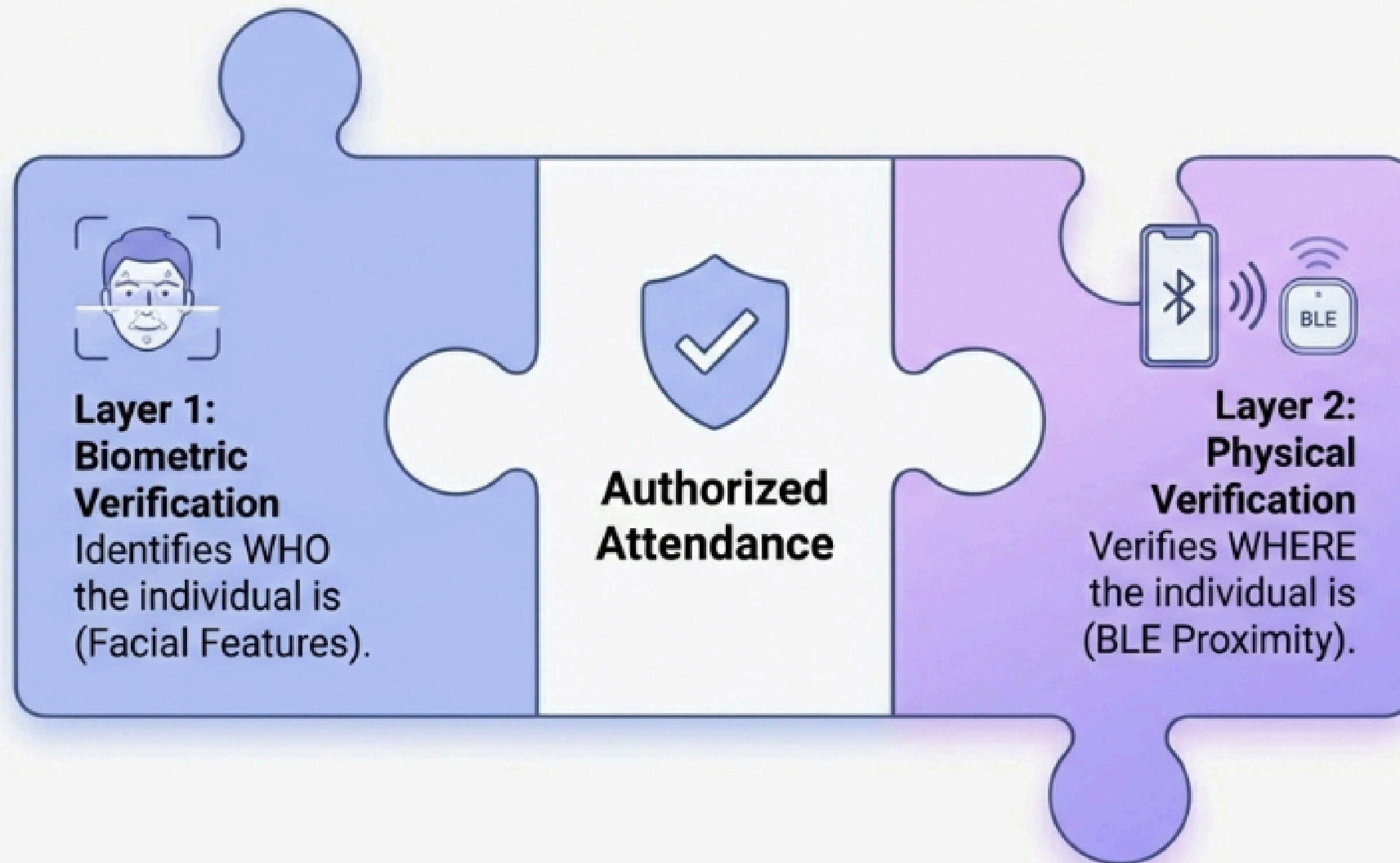
- ⚠ **Manual Registers:** Time-consuming and prone to manipulation.
- ⚠ **Standard QR Codes:** Insecure. Codes are easily photographed and shared remotely.
- ⚠ **Basic Face Scans:** Standard sensors are fooled by high-resolution photos.

The Impact



Core Problem: Proxy Attendance compromises institutional data integrity.

The Dual-Layer Security Architecture



The Key Differentiator: Attendance is marked only when the student is verified to be physically present in the classroom, effectively eliminating remote proxies.

Core Project Objectives



Elimination of Proxies

Mandating physical presence within BLE range.



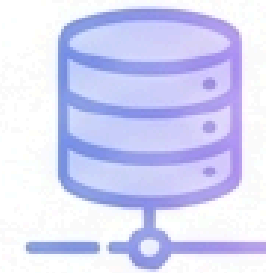
Liveness Assurance

Distinguishing real humans from photographs.



Time-Restricted Logging

Enforcing a 10-minute Validation Window.



Seamless Reporting

Automated database updates without manual entry.

Project Scope

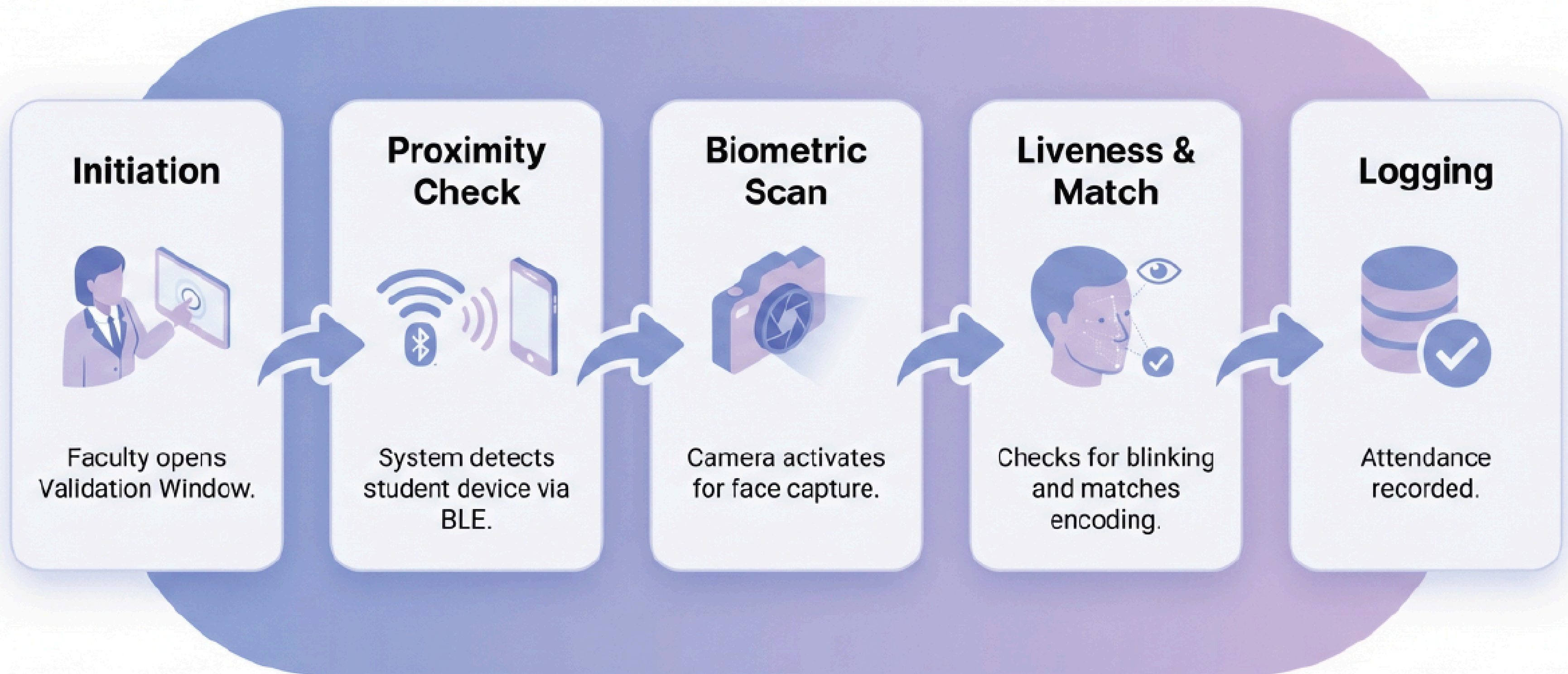
Current Scope:

- Colleges
- Universities
- Schools
- Training Institutes
- Corporate Training Sessions

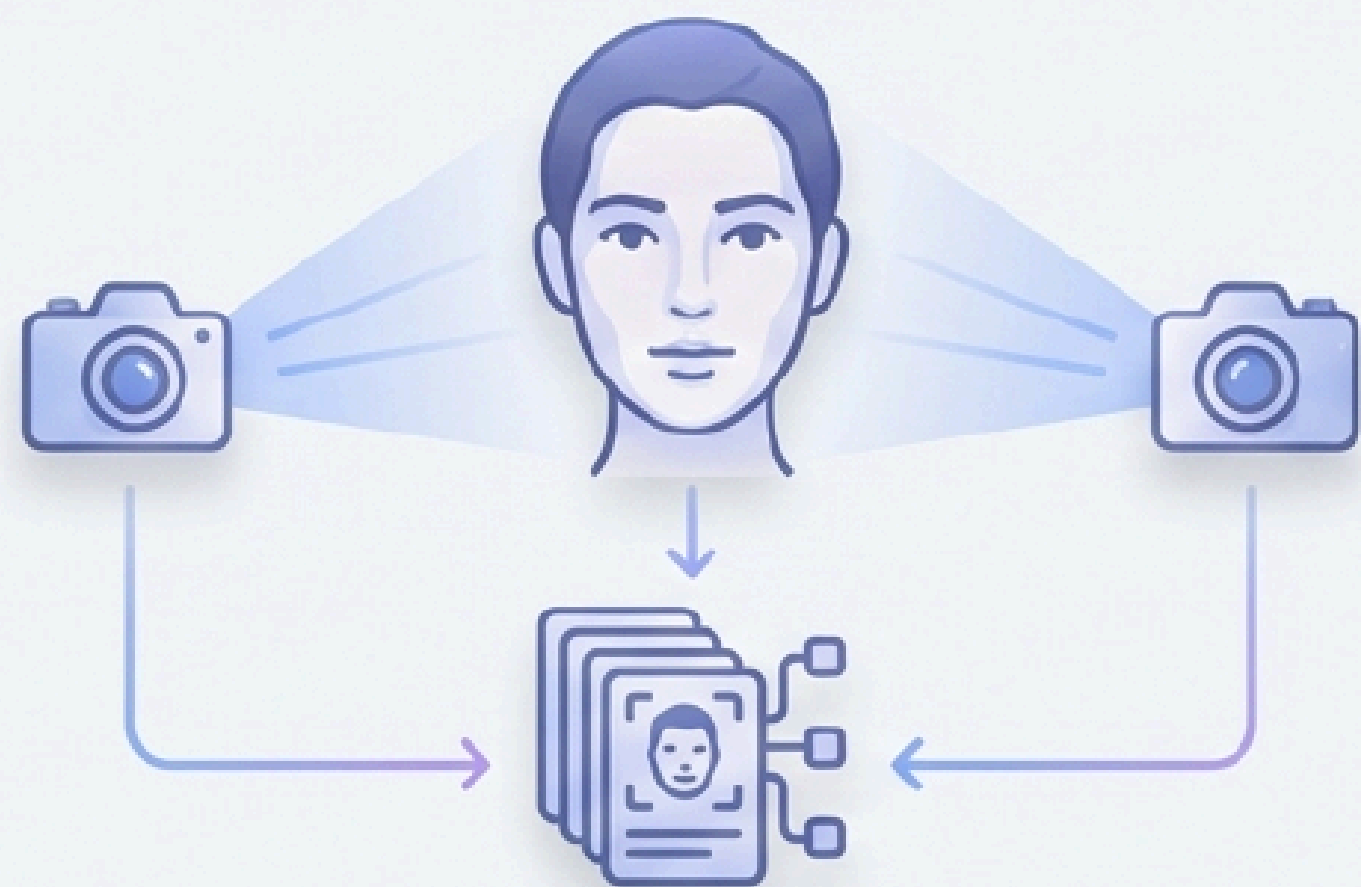
Future Scalability:

- Can integrate with ERP systems
- Cloud-based storage
- Mobile application integration

The Verification Workflow

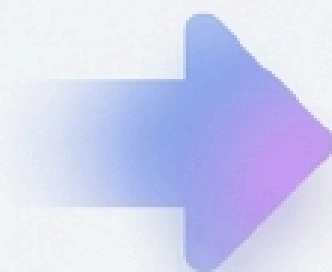


Module Deep Dive: Capture & Encoding



Module 1: Face Capture

Captures multiple images to account for angles and lighting. Organises raw imagery into a structured dataset.



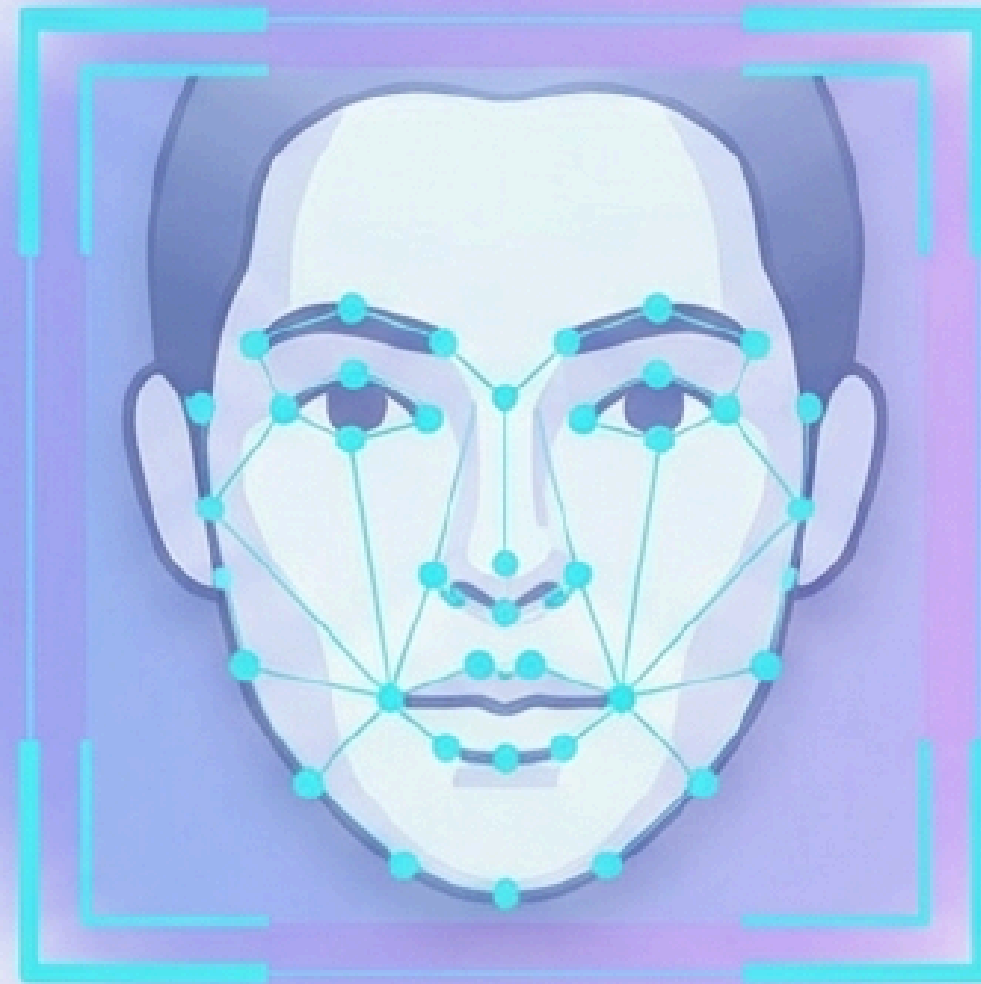
Module 2: Face Encoding

Converts raw pixel data into mathematical facial encodings. Saves unique signatures for recognition rather than raw images.

Module Deep Dive: Real-Time Detection

Module 3: Face Detection

Utilises Computer Vision to locate faces in video streams in real-time, isolating features from the background.



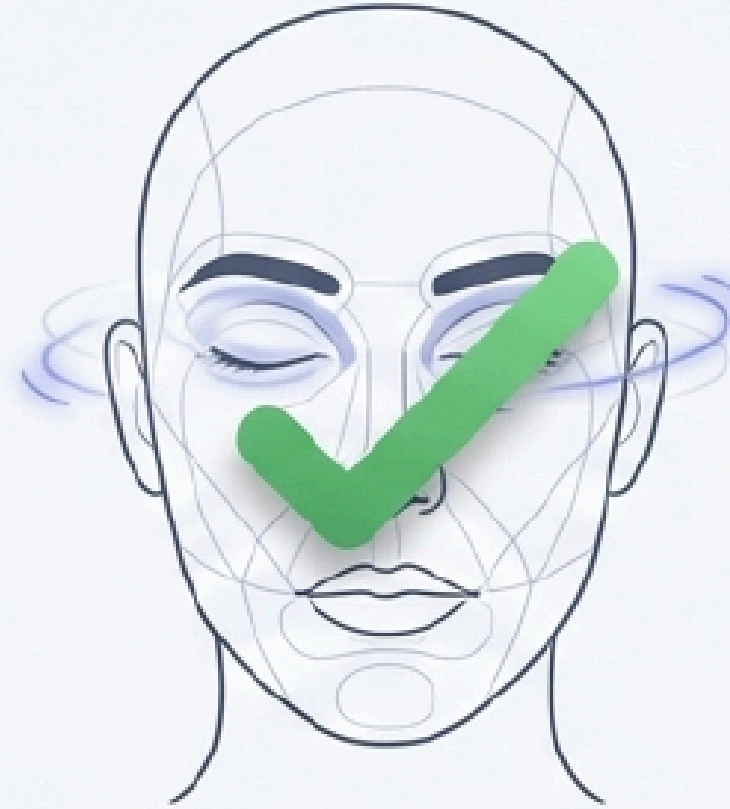
Module 4: Face Recognition

Compares the detected face against the database of pre-stored encodings. Logic prevents duplicate attendance entries.

Module Deep Dive : Liveness Detection



**Spoofing Attempt
(Static Image)**



**Verified Human
(Blink Detection)**

Module 5: Attendance Management

- **The Threat:** High-resolution photos used to spoof standard face scanners.
- **The Countermeasure:** Liveness Assurance tracks eye blinking to confirm the subject is a living, moving human.

Basic Requirements

Software and Hardware Requirements

Hardware Requirements

- Processor: Intel Core i3 (8th gen or higher) / AMD Ryzen 3 or equivalent
- RAM: 8 GB minimum (16 GB recommended for smoother real-time processing)
- Camera: HD webcam (720p minimum, 1080p preferred)
- BLE Device: BLE beacon or BLE-enabled smartphone
- Storage: 256 GB SSD minimum
- Server (local/cloud): Same minimum CPU/RAM as host or higher for scalability)

Software Requirements

- OS: Windows / Linux / macOS
- Programming: Python
- Libraries: OpenCV, Dlib, NumPy
- BLE communication library
- Database: MySQL
- Backend framework: Django
- Frontend: React
- Real-time data handling & API support
- Development Tools: VS Code
- Version Control: Git, Github



Literature Review

Existing System

- Uses webcam-based facial recognition (Haar, LBPH, dlib, CNN).
- Automates attendance but doesn't stop proxy misuse.
- Academic systems use small datasets and basic matching.
- Spoofing possible with photos or videos.
- Data often stored locally leading to security risks.
- Commercial solutions: eSSL, Mantra Softech, Secureye, SalaryBox, ZKTeco.
- Commercial systems are closed/proprietary with limited academic customization.



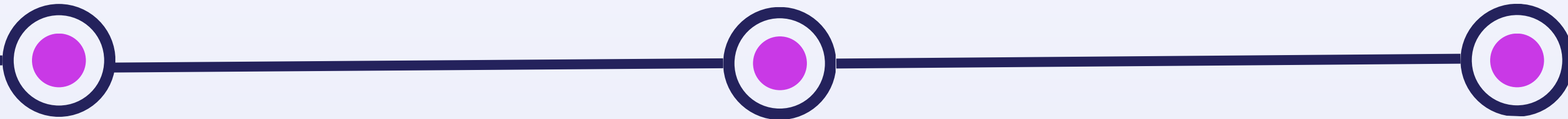
Literature Review

Proposed System

- Combines facial recognition + BLE location verification.
- Confirms physical presence at the correct location.
- Uses liveness detection (blink check) to stop spoofing.
- Stores secure face encodings, not raw images.
- Centralized real-time database for attendance.
- Scalable, secure, and proxy-resistant system.



Project Feasibility Study



Technical

- Python's Face_recognition library offers 99.38% accuracy on LFW dataset
- Works on standard system with webcam

Economical

- No expensive hardware required; utilizes existing student/faculty smartphones.
- Low implementation cost

Operational

- Zero friction for faculty; students only spend 5-10 seconds on verification.
- Reduces manual labor.
- Prevents Proxy attendance.

Requirement Gathering

- Literature Review: Studied existing face recognition and attendance systems. Reviewed online research papers and tutorials.
- Observation: Observed traditional attendance systems and their drawbacks.
- Online Research: Analyzed open-source face recognition projects and tools.
- User Feedback: Considered feedback from students and faculty regarding attendance issues.

Comparison with Existing System

Feature	Existing Systems	Proposed System
Attendance Method	Manual / Card-based	Face recognition
Time Consumption	High	Low
Accuracy	Prone to errors	High accuracy
Proxy Attendance	Possible	Reduced
Cost	Medium to High	Low
Automation	Partial	Fully automated
Anti-Spoofing	Not available	Basic spoof detection

Project Timeline Chart

ID	Task Name	Start	Finish	Duration	Dec 2025		Jan 2026				Feb 2026				Mar 2026			
					21-12	28-12	4-1	11-1	18-1	25-1	1-2	8-2	15-2	22-2	1-3	8-3	15-3	
1	Problem identification & requirement analysis	19-12-2025	26-12-2025	6d														
2	Literature review & feasibility study	24-12-2025	02-01-2026	8d														
3	Final project design & workflow	05-01-2026	09-01-2026	5d														
4	Environment setup & library installation	08-01-2026	12-01-2026	3d														
5	Face capture module development	13-01-2026	21-01-2026	7d														
6	Dataset creation & preprocessing	19-01-2026	27-01-2026	7d														
7	Face encoding module development	28-01-2026	05-02-2026	7d														
8	Exam break	02-02-2026	06-02-2026	5d														
9	Initial face recognition implementation	09-02-2026	13-02-2026	5d														
10	Attendance management system	16-02-2026	24-02-2026	7d														
11	Database integration	23-02-2026	27-02-2026	5d														
12	Anti-spoofing implementation	02-03-2026	06-03-2026	5d														
13	Performance optimization	09-03-2026	12-03-2026	4d														
14	System integration	13-03-2026	16-03-2026	2d														
15	Testing & debugging	17-03-2026	19-03-2026	3d														
16	Documentation	20-03-2026	20-03-2026	1d														
17	Final review & presentation	23-03-2026	23-03-2026	1d														

Project Work Distribution



23IT403

- Requirement gathering and analysis
- Face capture module development
- Dataset organization and preprocessing
- Attendance report generation
- Integration support and testing



23IT431

- System design and workflow planning
- Face encoding and model training
- Accuracy tuning and optimization
- Anti-spoofing feature implementation



23IT435

- Documentation preparation
- Camera handling and UI feedback
- Attendance database handling
- Error handling and validation
- System testing and debugging

Project Work Distribution

- Requirement gathering and analysis
- Face capture module development
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- Integration support and testing
- System design and workflow planning
- Face encoding and model training
- Accuracy tuning and optimization
- Anti-spoofing feature implementation
- Documentation preparation
- Camera handling and UI feedback
- Attendance database handling
- Error handling and validation
- System testing and debugging

The work is equally distributed among members of group

Documentation

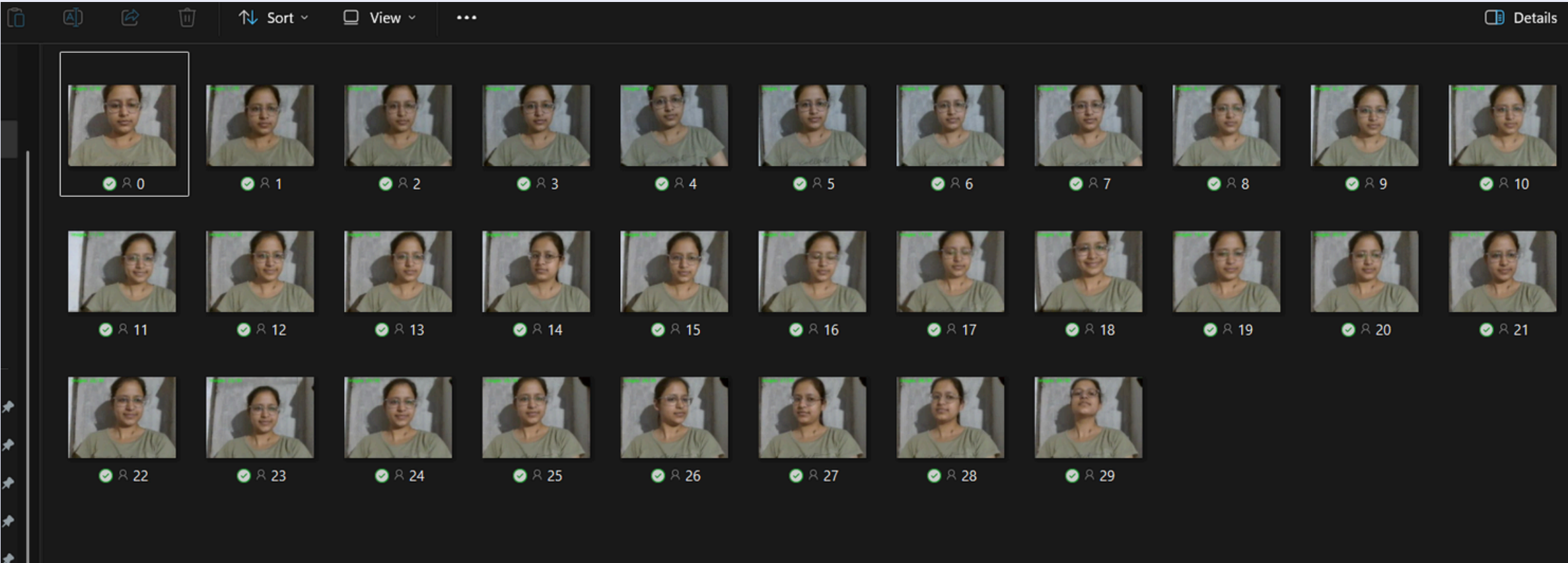
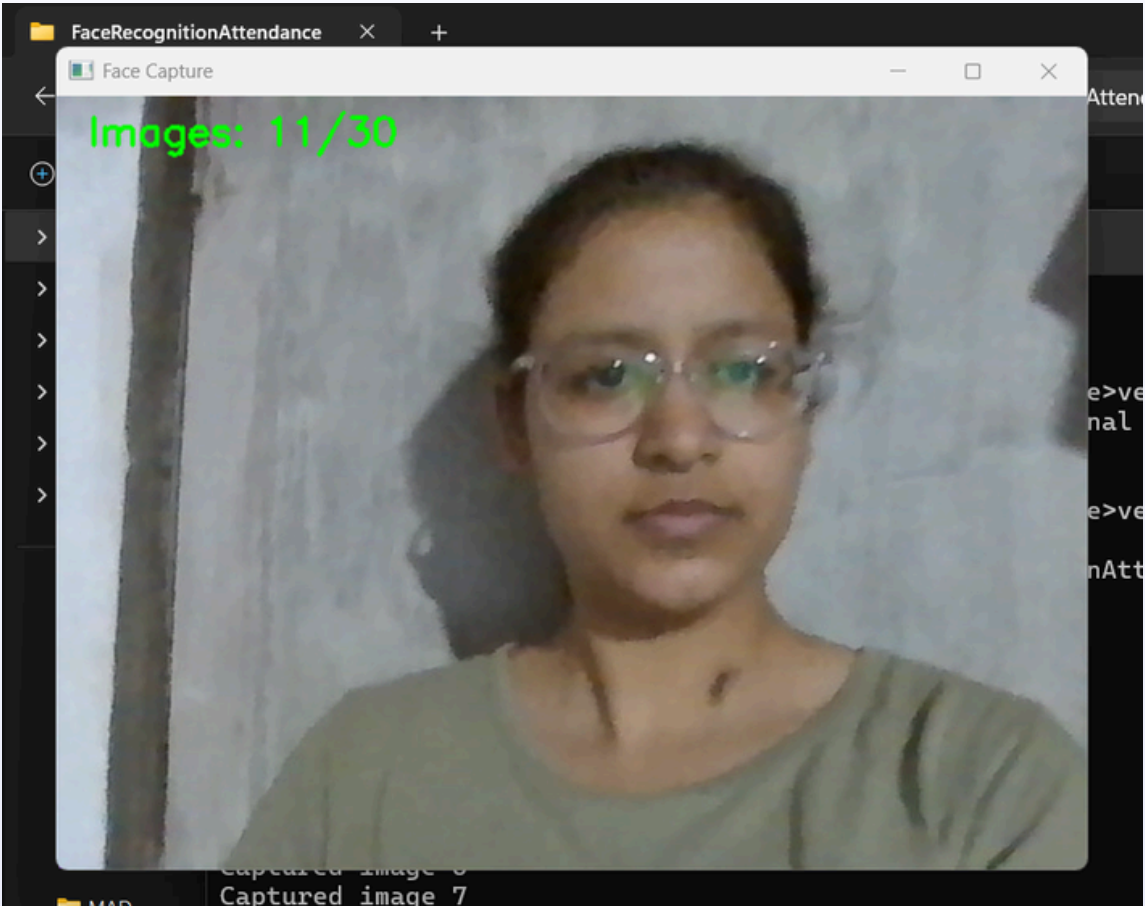
[Read the SRS](#)

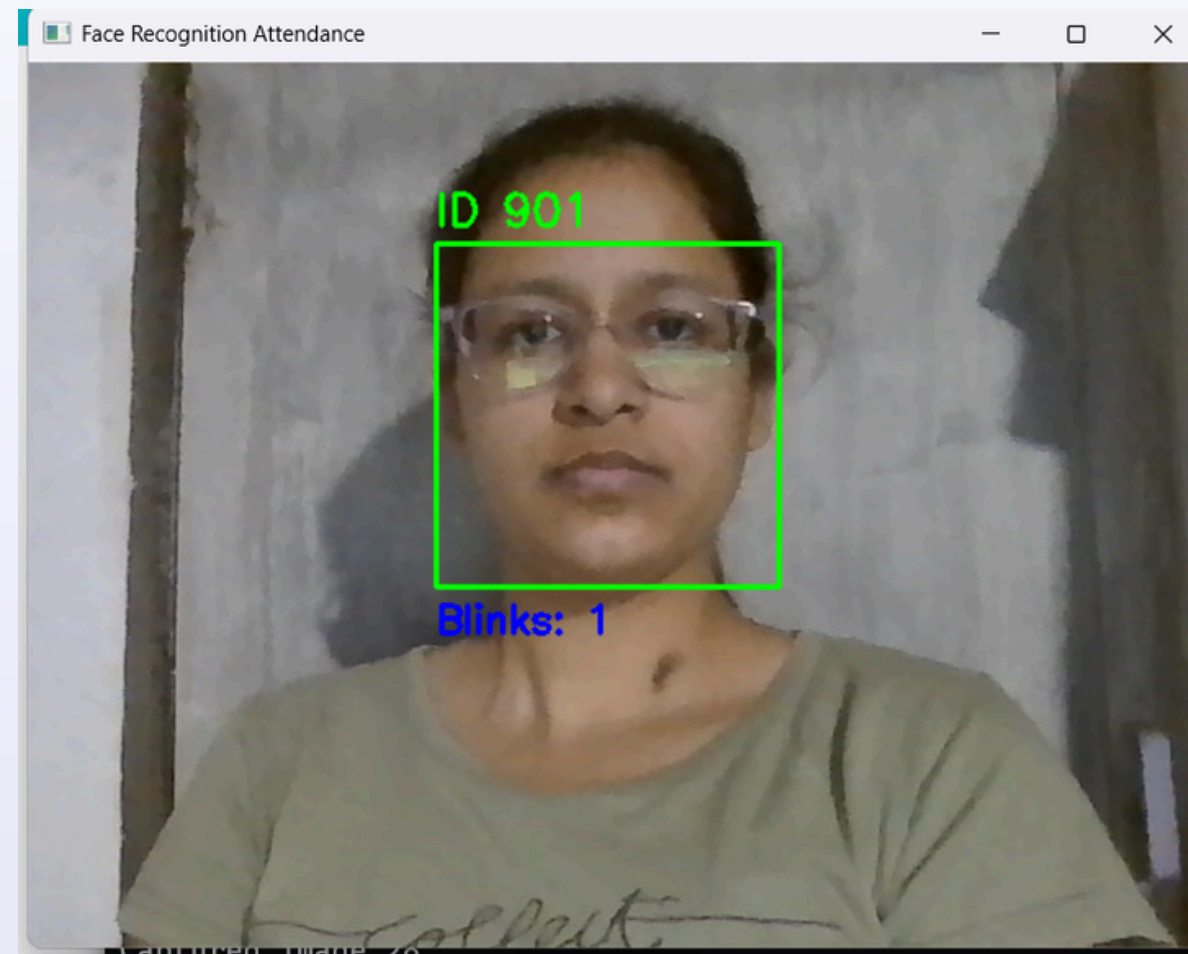
[Progress Report - 1](#)

Implementation

```
1  import cv2
2  import os
3  import time
4
5  # Ask for student ID
6  student_id = input("Enter Student ID: ")
7
8  # Create folder for student
9  # dataset_path = os.path.join("../", "dataset", f"student_{student_id}")
10 # os.makedirs(dataset_path, exist_ok=True)
11 BASE_DIR = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
12 dataset_path = os.path.join(BASE_DIR, "dataset", f"student_{student_id}")
13 os.makedirs(dataset_path, exist_ok=True)
14
15
16 # Open camera
17 cam = cv2.VideoCapture(0)
18
19 total_images = 30
20 count = 0
21 last_capture_time = time.time()
22
23 print("Camera started.")
24 print("Slowly move your face left, right, up, down.")
25
26 while True:
27     ret, frame = cam.read()
28     if not ret:
29         print("Failed to access camera")
30         break
31
```

Ln 15, Col 1 Spaces: 4 UTF-8 CRLF {} Python Signed out 3.10.11 (venv) Go Live





```
(venv310) C:\Users\veera\OneDrive\Desktop\FaceRecognitionAttendance>python src/recognize_attendance.py  
[INFO] Encodings loaded  
[INFO] Blink twice to mark attendance  
[ATTENDANCE] Marked for ID 901
```

THANK YOU